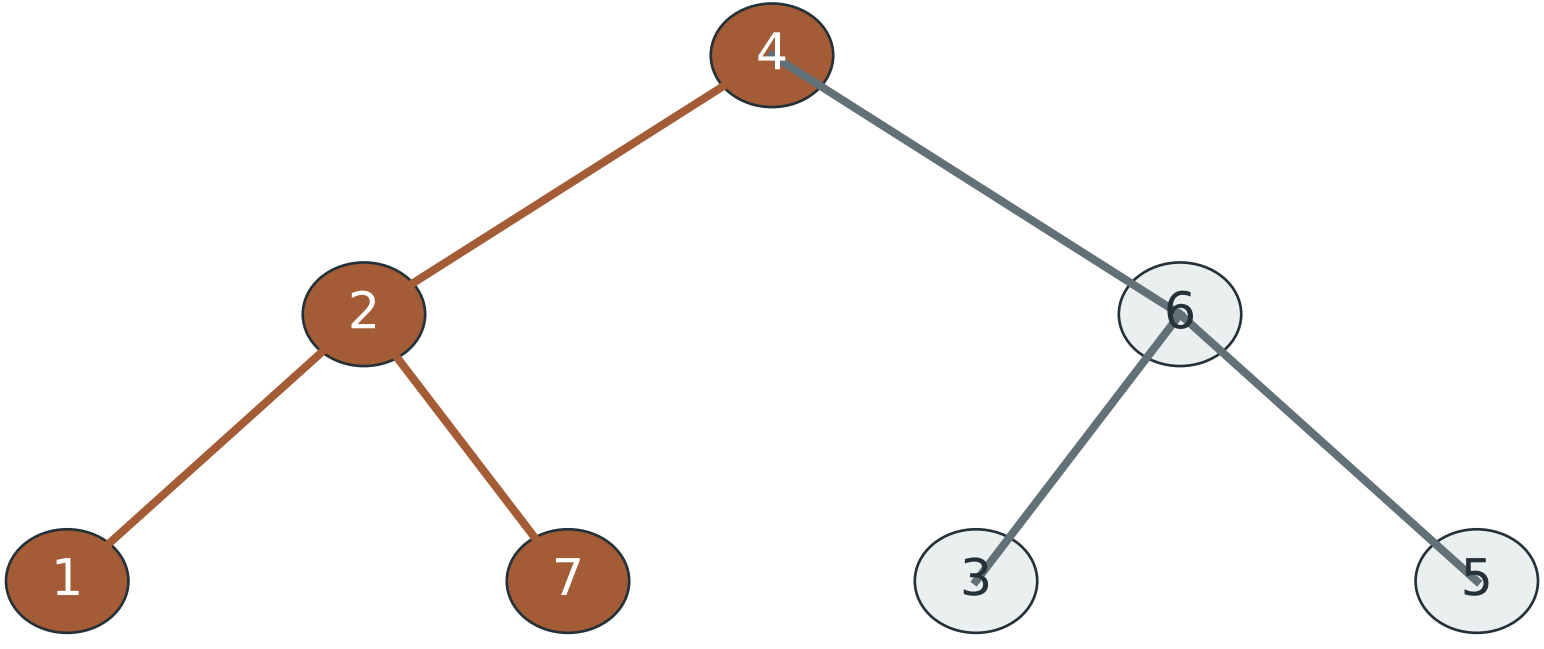

Selected star: four vertices, boundary size 4+2 = 6.